# SIMATS ENGINEERING
## ASSESSMENT TOOL1

**COURSE CODE: MLA 02 COURSE NAME: Fundamentals Of Machine Learning**

**COURSE OUTCOME COVERED**

***CO2:*** *Apply supervised learning algorithms for regression, classification, and predictive modeling tasks to solve structured data problems in practical applications. (BL3)*

*Assessment Tool Weightage: 40%*

**QUESTIONS: 10** **Total Marks:50**

Annexure A

**Analytical Problem Solving**

***Code of Conduct:***

*I B.Harsha Vardhan(192425201)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:B.HarshaVardhan***

Annexure A

## Analytical Problem Solving

Q1.A real estate company collected the following training data:(5Marks -10 Mins)

| House No | Area (sq.ft) | Bedrooms | Price (₹ Lakhs) |
|---|---|---|---|
| 1 | 1000 | 2 | 50 |
| 2 | 1500 | 3 | 75 |
| 3 | 800 | 2 | 40 |
| 4 | 1200 | 3 | 60 |
| 5 | 2000 | 4 | 90 |

(i) Construct a linear regression model of the form

$$Price = \beta_0 + \beta_1(Area) + \beta_2(Bedrooms)$$

The multiple linear regression model is

Price= β 0+ β 1 (Area) + β 2 (Bedrooms)

**where**

- **Price** = Predicted house price (Lakhs)
- **Area** = House area (sq.ft)
- **Bedrooms** = Number of bedrooms
- $\boldsymbol{\beta_0}$ = Intercept
- $\boldsymbol{\beta_1}$ = Area coefficient
- $\boldsymbol{\beta_2}$ = Bedroom coefficient

**(ii) Estimate the regression coefficients.**

Using the given dataset, the estimated regression equation is approximately

$$\boxed{Price=8.63+0.032(Area)+5.44(Bedrooms)}$$

- $\beta_0 = 8.63$
- $\beta_1 = 0.032$
- $\beta_2 = 5.44$

(iii) Interpret the meaning of each coefficient.

Intercept ($\beta_0 = 8.63$)

- It represents the estimated base price of a house when both area and bedrooms are zero.

- It is mainly a mathematical constant used by the regression model.

**Q2.An email system uses the following feature values:(5Marks -10 Mins)**

| Email | Word Frequency (Offer) | Word Frequency (Win) | Spam (1/0) |
|---|---|---|---|
| 1 | 3 | 2 | 1 |
| 2 | 2 | 1 | 1 |
| 3 | 1 | 0 | 0 |
| 4 | 3 | 3 | 1 |
| 5 | 0 | 1 | 0 |

**(i) Construct a linear classification model.**

The linear classification (Logistic Regression) model is

$g(x)=w0+w1X1+w2X2g(x)$

where

- $X1$= Word Frequency (Offer)
- $X2$ = Word Frequency (Win)

A suitable linear model for this dataset is

$\boxed{g(x)=-3+X1+X2}$

Decision Rule:

- If $g(x)\geq 0$ classify as **Spam (1)**
- If $g(x)<0$ classify as **Not Spam (0)**

**(ii) Determine the decision boundary.**

The decision boundary occurs when

$g(x)=0$

Substituting the model,

$-3+X1+X2=0$

Therefore,

$X1+X2=3$

This line separates the **Spam** and **Not Spam** emails.

(iii) Classify a new email with: Offer = 2, Win = 1.

For

- Offer = 2
- Win = 1

$g(x)=0$

**Predicted Class = Spam (1)**

Q3.A medical dataset is given:(5Marks -10 Mins)

| Patient | Fever (1/0) | Headache (1/0) | Disease (Yes/No) |
|---|---|---|---|
| 1 | 1 | 1 | Yes |
| 2 | 1 | 0 | Yes |
| 3 | 0 | 1 | No |
| 4 | 1 | 1 | Yes |
| 5 | 0 | 0 | No |

**(i) Compute prior probabilities.**

Disease = Yes

Number of patients with Disease = Yes = 3

$P(Disease=Yes)=35=0.6$

Disease = No

Number of patients with Disease = No = **2**

$P(Disease=No)$

| Disease | Probability |
|---|---|
| Yes | 3/5 = 0.6 |
| No | 2/5 = 0.4 |

**(ii) Compute conditional probabilities.**

Conditional Probability Table

| Conditional Probability | Value |
|---|---|
| P(Fever=1 \| Disease=Yes) | 1.0 |
| P(Fever=0 \| Disease=Yes) | 0.0 |
| P(Headache=1 \| Disease=Yes) | 2/3 = 0.667 |
| P(Headache=0 \| Disease=Yes) | 1/3 = 0.333 |
| P(Fever=1 \| Disease=No) | 0.0 |
| P(Fever=0 \| Disease=No) | 1.0 |
| P(Headache=1 \| Disease=No) | 0.5 |
| P(Headache=0 \| Disease=No) | 0.5 |

**(iii) Estimate the probability that a patient with Fever = 1 and Headache = 0 has the disease.**

Using Naïve Bayes:

For Disease = Yes

$$P(Yes)\times P(Fever=1 \mid Yes)\times P(Headache=0 \mid Yes)$$
$$=0.6\times 1\times 0.333$$
$$=0.20$$

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For Disease = No

$$P(No)\times P(Fever=1 \mid No)\times P(Headache=0 \mid No)$$
$$=0.4\times 0\times 0.5$$
$$=0$$

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Since

$0.20>0$

the patient is classified as

$\text{Disease} = \text{Yes}$

**Q4.Given the following dataset:(5Marks -10 Mins)**

| Data Point | $X_1$ | $X_2$ | Class |
|---|---|---|---|
| 1 | 2 | 3 | C1 |
| 2 | 3 | 4 | C1 |
| 3 | 5 | 6 | C2 |
| 4 | 6 | 7 | C2 |
| 5 | 4 | 5 | C2 |

**(i) Construct a discriminant function:**

$$g(x) = w_1x_1 + w_2x_2 + w_0$$

The linear discriminant function is

g(x)=w1X1+w2X2+w0

A simple discriminant function that separates the two classes is

g(x)=X1+X2−9

where

- w1=1
- w2=1
- w0=−9

**Decision Rule:**

- If g(x)<0, classify as **C1**
- If g(x)≥0, classify as **C2**

**(ii) Determine the decision boundary separating the two classes.**

The decision boundary is obtained by setting

g(x)=0

So,

X1+X2−9=0

Therefore,

$X1+X2=9$

This line separates **Class C1** and **Class C2**.

**(iii) Classify a new point ($X_1 = 3, X_2 = 3$).**

For (X1,X2)=(3,3):

$g(x)=3+3-9=-3$

Since g(x)<0,

Predicted Class = C1

**Q5.A telecom company collected the following data:(5Marks -10 Mins)**

| Customer | Monthly Charges | Tenure (Months) | Churn (1/0) |
|---|---|---|---|
| 1 | 500 | 12 | 0 |
| 2 | 700 | 8 | 1 |
| 3 | 400 | 15 | 0 |
| 4 | 900 | 5 | 1 |
| 5 | 600 | 10 | 1 |

**(i) Formulate the Bayesian logistic regression model.**

The logistic regression model is

$P(Y=1 \mid X)=1/1+e-Z1$

where

$Z=\beta 0+\beta 1X1+\beta 2X2$

Here,

- X1= Monthly Charges
- X2 = Tenure
- Y = Churn (1 = Yes, 0 = No)

Thus,

$P(Churn)=1/1+e-(\beta 0+\beta 1X1+\beta 2X2)$

**(ii) Estimate model parameters using the dataset.**

Using the given dataset, an approximate fitted model is

$Z = -8 + 0.01(\text{Monthly Charges}) - 0.30(\text{Tenure})$

Hence,

- Intercept ($\beta_0$) = -8
- Monthly Charges Coefficient ($\beta_1$) = 0.01
- Tenure Coefficient ($\beta_2$) = -0.30

## Interpretation

- Higher **Monthly Charges** increase the probability of churn.
- Higher **Tenure** decreases the probability of churn.

**(iii) Predict whether a customer with Monthly Charges = 650 and Tenure = 9 will churn.**

For

- Monthly Charges = **650**
- Tenure = **9**

$P(\text{Churn}) \approx 0.015P$

Since $P<0.5$,

$\text{Predicted Class} = \text{No Churn} \ (0)$

**Q6.Given the following dataset for customer classification:(5Marks -10 Mins)**

| Customer | Age | Income | Class |
|---|---|---|---|
| C1 | 25 | 40 | A |
| C2 | 30 | 60 | A |
| C3 | 35 | 80 | B |
| C4 | 40 | 20 | B |

**For a new customer (Age = 32, Income = 50):**

**(i) .Compute Euclidean distance from all points**

The Euclidean distance formula is

d=(x1 −x2 )2+(y1 −y2 )2

Distance from C1 (25,40)

d=(32−25)2+(50−40)2 =72+102 =49+100 =149 =12.21

**(ii).Classify using k = 3**

Arrange distances in ascending order:

| Nearest Neighbor | Distance | Class |
|---|---|---|
| C2 | 10.20 | A |
| C1 | 12.21 | A |
| C3 | 30.15 | B |

Among the **3 nearest neighbors**:

- Class **A = 2**
- Class **B = 1**

Majority class is **A**.

**(iii) .Determine the final class label**

**Q7.Given the dataset:(5Marks -10 Mins)**

| **Outlook** | **Temperature** | **Play** |
|---|---|---|
| **Sunny** | **Hot** | **No** |
| **Sunny** | **Mild** | **No** |
| **Overcast** | **Hot** | **Yes** |
| **Rain** | **Mild** | **Yes** |

**(i) . Compute Entropy of Play**

Formula:

$Entropy(S)=-P(Yes)\log2 \quad P(Yes)-P(No)\log2$

Here,

- Yes = 2
- No = 2
- Total = 4

Therefore,

$P(Yes)=42 \quad =0.5 \ P(No)=42 \quad =0.5 \ Entropy(S)=-(0.5)\log2 \quad (0.5)-(0.5)\log2 \quad (0.5) =-0.5(-1)-0.5(-1) \ =0.5+0.5 \ Entropy(S)=1$

**(ii). Calculate Information Gain for attribute Outlook**

Information Gain

$IG(Outlook)=Entropy(S)-WeightedEntropy \ =1-0 \ IG(Outlook)=1$

**(iii). Identify the root node of the decision tree**

The attribute having the highest Information Gain becomes the root node.

Since

$IG(Outlook)=1 \ Root \quad Node \quad = \quad Outlook$

**Q8.Given data points:(5Marks -10 Mins)**

| X1 | X2 | Class |
|---|---|---|
| 2 | 2 | +1 |
| 4 | 4 | +1 |
| 2 | 4 | -1 |
| 4 | 2 | -1 |

**(i) .Plot the points**

Positive (+1)

- (2,2)
- (4,4)

Negative (-1)

- (2,4)
- (4,2

**(ii).Determine a linear separating hyperplane**

One possible separating hyperplane is

$x1 \; - x2 \; = 0$

**(iii) .Find the equation of the decision boundary**

Therefore,

$x1 \; = x2$

**Q9.Given confusion matrices:(5Marks -10 Mins)**

**Logistic Regression:**

| | Pred + | Pred - |
|---|---|---|
| Actual + | 40 | 10 |
| Actual - | 5 | 45 |

**Naïve Bayes:**

| | Pred + | Pred - |
|---|---|---|
| Actual + | 35 | 15 |
| Actual - | 10 | 40 |

**(i) .Compute Accuracy, Precision, Recall for both models**

Confusion Matrix

| | Pred + | Pred − |
|---|---|---|
| Actual + | 40 | 10 |
| Actual − | 5 | 45 |

### Accuracy

Accuracy=TotalTP+TN =10040+45 =0.85 Accuracy=85%

**(ii).Compare results and identify the better model**

## Comparison

| Metric | Logistic Regression | Naïve Bayes |
|---|---|---|
| Accuracy | 85% | 75% |
| Precision | 88.89% | 77.78% |
| Recall | 80% | 70% |

**Q10.Given points:(5Marks -10 Mins)**

**(2, 4), (3, 5), (8, 9), (9, 10)**

**Initial centroids:**

**C1 = (2,4), C2 = (8,9)**

**(i) .Perform one iteration of K-means**

| Point | Distance to C1 | Distance to C2 | Cluster |
|---|---|---|---|
| (2, 4) | 0 | 7.81 | C1 |
| (3, 5) | 1.41 | 6.40 | C1 |
| (8, 9) | 7.81 | 0 | C2 |
| (9, 10) | 9.22 | 1.41 | C2 |

**(ii) Cluster Assignment**

Cluster 1

- (2,4)
- (3,5)

Cluster 2

- (8,9)
- (9,10)

(iii) Compute New Centroids

## Cluster 1

(22+3 , 24+5 ) =(2.5, 4.5)

## Cluster 2

(28+9 , 29+10 ) =(8.5, 9.5)